ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 3

MECHATRONICS

PAPER 1 · PRACTICE C · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 3 Mechatronics paper 1 · practice c.

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How to mark: each ✓ is one mark. Give marks for the formula, the substitution and the answer with the correct unit. Drawings: award marks only for components that are drawn with the correct symbol and connected correctly. Accept other correct answers.

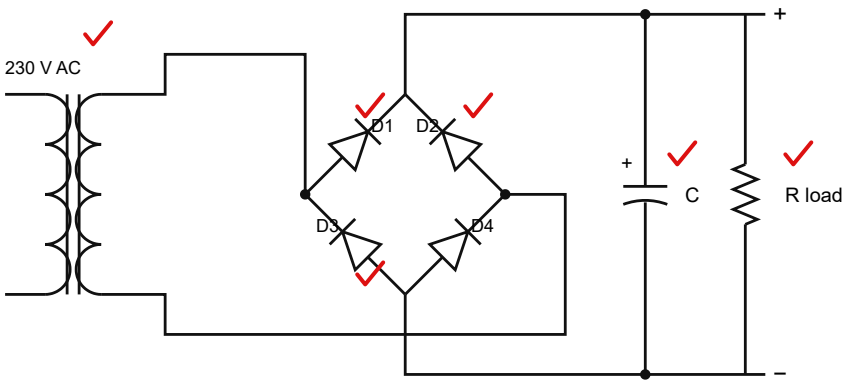
QUESTION 1

1.1 A resistor is a component that opposes (limits) the flow of current ✓ in a circuit, measured in ohms ✓. (2)

1.2 Brown = 1, black = 0, red = × 100 ✓ → 1 000 Ω (1 kΩ) ✓; gold = ±5% tolerance ✓ (3)

1.3 AC changes direction (and size) continuously, e.g. the mains supply. ✓✓ (4)
DC flows in one direction only at a steady value, e.g. from a battery. ✓✓ (2 × 2)

1.4



(Transformer 1; four diodes in a bridge, all facing the correct way 2; smoothing capacitor across the output 1; load resistor 1; labels and polarity 1) (6)

1.5.1 $R_T = (R_1 \times R_2) \div (R_1 + R_2)$ ✓ $= (100 \times 150) \div 250$ ✓ $= 60\ \Omega$ ✓ (3)

1.5.2 $I = V \div R = 12 \div 60$ ✓ $= 0{,}2\ A$ ✓ (2)

[20]

QUESTION 2

2.1 A servo motor is a motor with a built-in feedback device (e.g. a potentiometer or encoder) ✓ that moves to and holds an exact position (angle) set by a control signal ✓. (2)

2.2 By reversing the polarity of the supply to the armature ✓ (or to the field winding, but not both) ✓. (2)

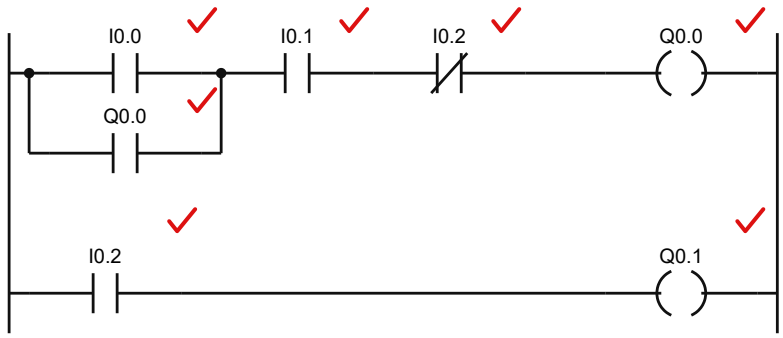
2.3 Any THREE ✓✓✓: armature (rotor); field windings / permanent magnets (stator); commutator; brushes; shaft and bearings; yoke (frame). (3)

2.4.1 Q0.0 switches on and stays on after I0.0 is released ✓; Q0.1 also switches on ✓. (2)

2.4.2 It is a holding (latching) contact ✓ that keeps Q0.0 on after the START button is released ✓. (2)

2.4.3 The I0.1 input opens (it is wired NC) ✓, so Q0.0 switches off and the latch is released; Q0.1 also switches off ✓. (2)

2.5



I0.2 = float switch (closes when the tank is full). Q0.1 = "tank full" lamp.

(I0.0 NO 1 with a Q0.0 holding contact in parallel 1; I0.1 programmed NO (closed in normal operation) 1; float switch I0.2 as an NC contact in series 1; output coil Q0.0 1; second rung: I0.2 NO driving the lamp Q0.1 1; power rails and labels 1) (7)

[20]

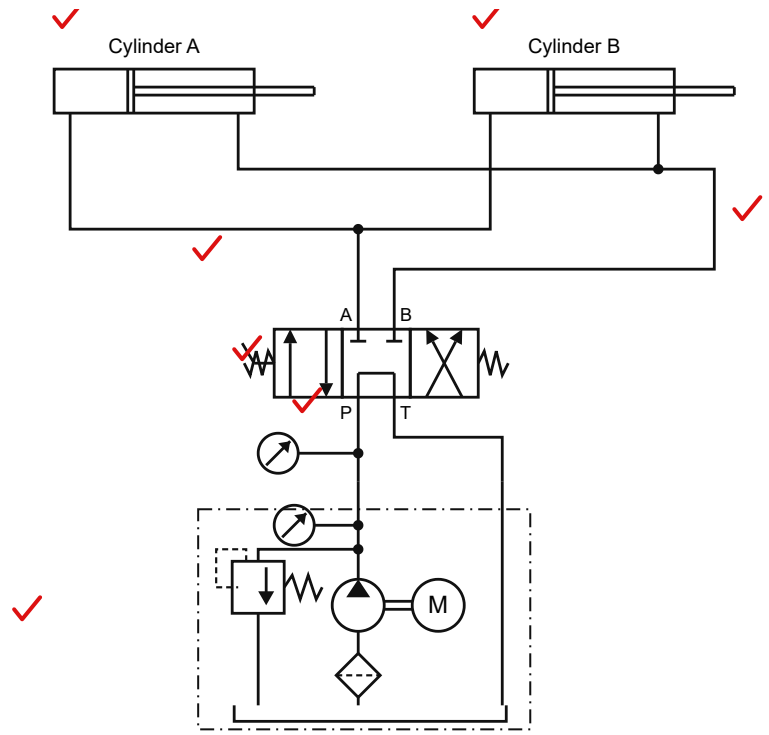
QUESTION 3

3.1 A filter removes dirt and particles from the oil ✓ to protect the pump, valves and seals. Any TWO places ✓✓: suction line (strainer); pressure line; return line to the tank. (3)

3.2.1 $A = \pi \times d^2 \div 4 = \pi \times 0{,}08^2 \div 4 = 5{,}027 \times 10^{-3}\ m^2$ ✓; $F = p \times A = 10 \times 10^6 \times 5{,}027 \times 10^{-3}$ ✓ $= 50\ 265\ N\ (50{,}27\ kN)$ ✓ (3)

3.2.2 $A = \pi \times (D^2 - d^2) \div 4 = \pi \times (0{,}08^2 - 0{,}04^2) \div 4 = 3{,}770 \times 10^{-3}\ m^2$ ✓; $F = 10 \times 10^6 \times 3{,}770 \times 10^{-3}$ ✓ $= 37\ 699\ N\ (37{,}70\ kN)$ ✓ (3)

3.3



(Two double-acting cylinders 2; cap ends joined to port A 2; rod ends joined to port B 2; 4/3 lever valve with a tandem centre 2; power pack with relief valve 2; pressure gauge 1)

(11)

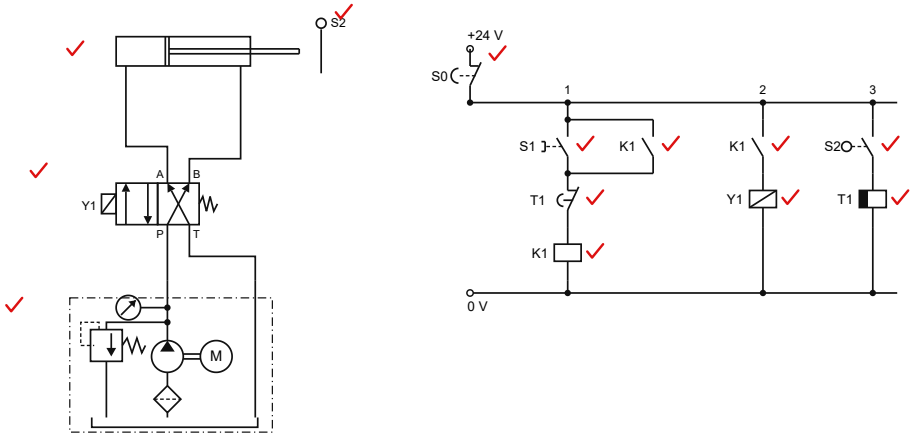
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QUESTION 4

4.1 It detects the position of the cylinder (e.g. fully extended) ✓ and sends a signal to the control circuit to start the next step or stop the movement ✓. (2)

4.2 A direct-operated solenoid valve: the solenoid moves the spool directly, used for small valves and low flow. ✓✓ (4)
A pilot-operated solenoid valve: the solenoid moves a small pilot valve, and the oil pressure then moves the main spool – used for large valves with high flow. ✓✓ (2 × 2)

4.3



(Working circuit: power pack 1; 4/2 single-solenoid valve with spring return 1; cylinder correctly connected 1; limit switch S2 at full extension 1. Control circuit: supply lines 1; E-stop (NC) in the supply 1; S1 (NO) 1 with K1 holding contact in parallel 1; NC T1 contact in series with the K1 coil 1; K1 coil 1; K1 contact switching Y1 1; S2 (NO) switching the timer T1 1 set to 3 s 1; correct symbols and neat 1)

(14)

[20]

QUESTION 5

5.1 Maintenance done when measurements of the machine's condition (vibration, temperature, oil analysis) show that a part is starting to fail ✓, so it is repaired before it breaks down ✓. (2)

5.2 (4)
- Connect the sensor to its supply (e.g. 24 V) ✓
- Set the multimeter to DC volts and measure between the output wire and 0 V ✓
- Without a target the output should be about 0 V ✓
- With a metal target in front of the sensor the output should rise to about the supply voltage (and the LED should light) ✓

5.3 Any FIVE ✓✓✓✓✓: oil level too low; relief valve set too low so oil keeps flowing over it; blocked or dirty cooler / reservoir; wrong oil viscosity; dirty or blocked filter; internal leaks in worn pumps or cylinders; system working at high pressure for too long; high room temperature. (5)

5.4 (5)
- Tell the people affected that the machine will be shut down ✓
- Switch off the machine with its normal controls ✓
- Isolate the energy sources (electrical isolator, hydraulic pressure released) ✓
- Fit your own lock and a tag to the isolator ✓
- Test that the machine cannot start (try the START button) before working ✓

5.5 Any FOUR ✓✓✓✓: plan future maintenance; show how often a part fails; know when parts were replaced (warranty); control costs and order spares; legal / safety requirement; help with fault-finding. (4)

[20]

TOTAL: 100

END OF MARKING GUIDELINE